Exercise Sheet

Lecture 14

Operators on Hilbert Space

$\mathrm{spec}(T)$ is the spectrum; $T^\dagger$ the adjoint; inner products follow the physics convention. A $\star$ marks a harder problem.

Bounded operators, adjoints, spectrum

Exercise 14.1 (computational) Find the operator norm and the adjoint of multiplication by $m(x) = e^{ix}$ on $L^2[0,2\pi]$.

Exercise 14.2 (computational) The diagonal operator $De_n = \frac{n}{n+1}e_n$ on ℓ^2. Find its point spectrum and full spectrum.

Exercise 14.3 (computational) Let $De_n = 2^{-n}e_n$ on ℓ^2, where $e_1, e_2, \ldots$ is the standard orthonormal basis ($n \geq 1$). Find $\|D\|$, the adjoint $D^\dagger$, the point spectrum, and the full spectrum; is D self-adjoint? compact?

Exercise 14.4 (computational) Find the operator norm, the adjoint, and the spectrum of multiplication by $g(x) = 2x$ on $L^2[0,1]$. Does it have eigenvalues?

Exercise 14.5 (computational) Let S be the right shift $S(x_1, x_2, \ldots) = (0, x_1, x_2, \ldots)$ on ℓ^2. Compute $S^\dagger S$ and $SS^\dagger$, give $\|S\|$, and decide whether S is an isometry and whether it is unitary.

Exercise 14.6 (computational) For $A = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on $\mathbb{C}^2$, find the adjoint $A^\dagger$, classify A (self-adjoint / unitary / normal), and give $\mathrm{spec}(A)$.

Exercise 14.7 ($\star$) Show that multiplication by x on $L^2[0,1]$ has no eigenvalues, yet $\mathrm{spec}(\hat{x}) = [0,1]$.

Exercise 14.8 (proof) Show that a bounded diagonal operator $De_n = d_n e_n$ on ℓ^2 (with $\sup_n |d_n| < \infty$) has $\|D\| = \sup_n |d_n|$.

Position, momentum, and the commutator

Exercise 14.9 (proof) Verify $[\hat{x}, \hat{p}]\psi = i\hbar\psi$ for $\hat{p} = -i\hbar\, d/dx$ on a smooth ψ.

Exercise 14.10 (proof) Prove no finite matrices X, P satisfy $[X, P] = i\hbar I$.

Exercise 14.11 (computational) Using $[A, BC] = [A,B]C + B[A,C]$, compute $[\hat{x}, \hat{p}^2]$.

Exercise 14.12 (computational) Using $[A, BC] = [A,B]C + B[A,C]$ and $[\hat{x}, \hat{p}] = i\hbar$, compute $[\hat{p}, \hat{x}^2]$.

Exercise 14.13 (computational) Compute $[\hat{x}, \hat{p}^3]$.

Exercise 14.14 (computational) Compute $[\hat{x}^2, \hat{p}^2]$, leaving the answer in terms of $\hat{x}$ and $\hat{p}$.

Exercise 14.15 (computational) In natural units ($\hbar = 1$, so $[\hat{x}, \hat{p}] = i$) set $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$ and $a^\dagger = \frac{1}{\sqrt{2}}(\hat{x} - i\hat{p})$. Compute $[a, a^\dagger]$.

Exercise 14.16 (computational) With $a, a^\dagger$ as above and the number operator $N = a^\dagger a$, use $[a, a^\dagger] = \mathrm{id}$ to compute $[N, a]$.

Exercise 14.17 (conceptual) *(The boundary condition is part of the observable.)* Let $T = -\dfrac{d^2}{dx^2}$ act on twice-differentiable functions on $[0, 1]$ with the L^2 inner product.

1. Integrating by parts twice, show that
$$\langle Tu, v\rangle - \langle u, Tv\rangle = \Big[\overline{u}\, v' - \overline{u'}\, v\Big]_0^1.$$
2. Conclude that T is symmetric ($\langle Tu, v\rangle = \langle u, Tv\rangle$) on the Dirichlet domain $u(0) = u(1) = 0$, and explain why the same operator with *no* boundary condition is not.

Exercise 14.18 (proof) Prove $[\hat{x}, \hat{p}^{\,n}] = n\, i\hbar\, \hat{p}^{\,n-1}$ for all $n \geq 1$ by induction.

The spectral theorem and dynamics

Exercise 14.19 (computational) For the compact self-adjoint T with $Te_n = \frac{1}{n} e_n$ on ℓ^2, write Tv and $\|Tv\|^2$ for $v = \sum_n c_n e_n$.

Exercise 14.20 (computational) For $H = \frac{\hbar\omega}{2}\sigma_z$, evolve $|\psi\rangle(0) = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ to time t.

Exercise 14.21 (computational) Measuring σ_x on $|0\rangle$: give the outcome probabilities and $\langle\sigma_x\rangle$.

Exercise 14.22 (computational) Let $Te_n = \frac{1}{n} e_n$ on ℓ^2 (compact, self-adjoint) and $v = 3e_1 + 4e_2$. Compute Tv, $\|v\|^2$, $\|Tv\|^2$, $\langle v, Tv\rangle$, and the Rayleigh quotient $\langle v, Tv\rangle / \|v\|^2$.

Exercise 14.23 (computational) A three-level system has $H|n\rangle = n\varepsilon|n\rangle$ for $n = 0, 1, 2$. Evolve $|\psi\rangle(0) = \frac{1}{\sqrt{3}}(|0\rangle + |1\rangle + |2\rangle)$ to time t.

Exercise 14.24 (computational) Measuring σ_z in $|\psi\rangle = \frac{1}{2}(\sqrt{3}\,|0\rangle + |1\rangle)$: give the outcome probabilities and $\langle\sigma_z\rangle$.

Exercise 14.25 (computational) Using the spectral decomposition $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$, compute $e^{i\alpha\sigma_z}$ for real α.

Exercise 14.26 ($\star$) For a density operator ρ, prove $\langle A\rangle = \mathrm{tr}(\rho A)$ reduces to $\langle\psi, A\psi\rangle$ when $\rho = |\psi\rangle\langle\psi|$ is pure.

Exercise 14.27 (proof) For a self-adjoint $H = \sum_n E_n |e_n\rangle\langle e_n|$ with real E_n and orthonormal $\{e_n\}$, show that $U(t) = e^{-iHt/\hbar}$ acts by $|e_n\rangle \mapsto e^{-iE_n t/\hbar}|e_n\rangle$ and is unitary.

Perturbation in infinite dimensions

Exercise 14.28 (computational) Let D be the diagonal operator $De_n = \frac{1}{n} e_n$ on ℓ^2 and let $z = -1$. Compute $\|(D - z\,\mathrm{id})^{-1}\|$, and find the largest $|\varepsilon|$ for which the resolvent expansion $(D + \varepsilon V - z\,\mathrm{id})^{-1} = \sum_{k \geq 0} (-\varepsilon)^k R_0(z) \big(V R_0(z)\big)^k$ is guaranteed to converge, for a bounded self-adjoint V with $\|V\| = 2$.

Exercise 14.29 ($\star$) Explain why the Rayleigh–Schrödinger series for the quartic anharmonic oscillator $H(\varepsilon) = \frac{1}{2}(\hat{p}^2 + \hat{x}^2) + \varepsilon \hat{x}^4$ cannot have a positive radius of convergence, even though its first few terms are numerically excellent. Contrast this with the finite-dimensional situation of Lecture 10.